

## Cheng Zhi Min A25CS0050 Assignment 2

1.

4000 0000h 0A

1 14

2 1E

3 E8

4 03

5 F4

6 01

7 00

8 00

9 50

A C3

B 00

C 00

D 10

E 27

F 00

10 00

myBytes

myWords

myDwords

i. ESI = 4000 0000h

ii. EDI = 4000 0003h

iii. CL = 0Ah

iv. BX = 1E14h

v. EDX = 0001 F403h

vi. BX = 01F4h

Line 14: register indirect

Line 18: immediate

Line 19: register indirect

Line 22: indexed.

2. 3000h 69

1 00

2 0B

3 00

4 6A

5 00

6 0F

7 ?

8 5A

9 5A

A 43

B 41

C 52

setArr1

setArr2

i. AX = 000Bh

ii. DH = 5Ah

iii. BX = 6A00h

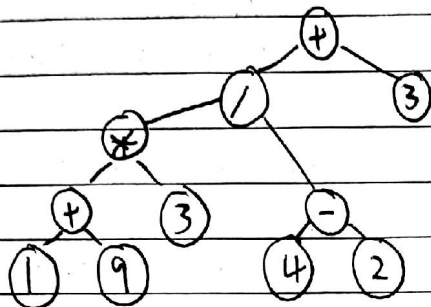
iv. DL = 43h

(c) Yes. Dumpregs is only used to display register contents for output.

$$4. \text{ ii. } M = ((1+9)*3)/(4-2)+3 \Rightarrow 19 + 3*42 - /3 +$$

$$19 \xrightarrow{((+)} 19+3 \xrightarrow{(*)} 19+3*42 \xrightarrow{/( - ) +} 19+3*42- / \xrightarrow{+} 19+3*42- /3 +$$

i.



| +  | *  | 2  | /  | +  |
|----|----|----|----|----|
| 9  | 3  | 4  | 2  | 3  |
| 1  | 10 | 30 | 30 | 15 |
| 18 |    |    |    |    |
| ①  | ②  | ③  | ④  | ⑤  |
|    |    |    |    | ⑥  |

$$M=18$$

(c) i. LOAD 4  
SUB 2  
STO T  
LOAD 1  
ADD 9  
MUL 3  
DIV T  
ADD 3  
STO Q

ii. PUSH 1  
PUSH 9  
ADD  
PUSH 3  
MUL  
PUSH 4  
PUSH 2  
SUB  
DIV  
PUSH 3  
ADD  
POP Q

|                    |               |             |           |        |
|--------------------|---------------|-------------|-----------|--------|
| 3. i. ADD T1, A, B | ii. MOV R1, A | iii. LOAD F | iv. PUSHA | PUSH G |
| MUL T1, T1, C      | ADD R1, B     | MUL G       | PUSH B    | MUL    |
| SUB T1, T1, D      | MUL R1, C     | ADD E       | ADD       | ADD    |
| MUL T2, F, G       | SUB R1, D     | STORE T     | PUSH C    | DIV    |
| ADD T2, E, T2      | MOV R2, F     | LOAD A      | MUL       | POP Q  |
| DIV Q, T1, T2      | MUL R2, G     | ADD B       | PUSH D    |        |
|                    | ADD R2, E     | MUL C       | SUB       |        |
|                    | DIV R1, R2    | SUB D       | PUSH E    |        |
|                    |               | DIV T       | PUSH F    |        |
|                    |               | STORE Q     |           |        |